Time Series Analysis & Classification

T. Medkour & H. Brairi

National School of Artificial Intelligence

Chapter II: Modeling Deterministic Trends

1. Introduction

2. Estimation of a constant mean

3. Regression methods

4. Interpreting Regression Output

5. Residual Analysis (Model Diagnostics)

Introduction

Discussion

- In this course, we consider time series models for realizations of a stochastic process $\{ Y_t : t = 0, 1, \ldots, n\}$. This will largely center around models for stationary processes. However, as we have seen, many time series data sets exhibit a trend; i.e., a long-term change in the mean level. We know that such series are not stationary because the mean changes with time.

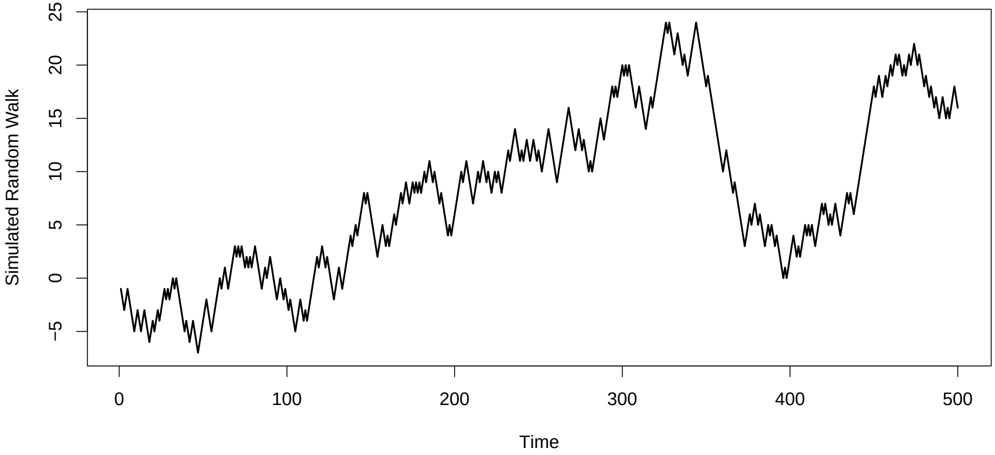
Discussion

- An obvious difficulty with the definition of a trend is deciding what is meant by the phrase long-term. For example, climatic processes can display cyclical variation over a long period of time, say, 1000 years. However, if one has just 40-50 years of data, this long-term cyclical pattern might be missed and be interpreted as a trend which is linear.

Discussion

- Trends can be elusive, and an analyst may mistakenly conjecture that a trend exists when it really does not. For example, in a realization of a random walk process $Y_t = Y_{t-1} + e_t$, where $e_t \sim N(0, 1)$.

Discussion



Simulated Random Walk
Time
−5
0
5
10
15
20
25
0
100
200
300
400
500

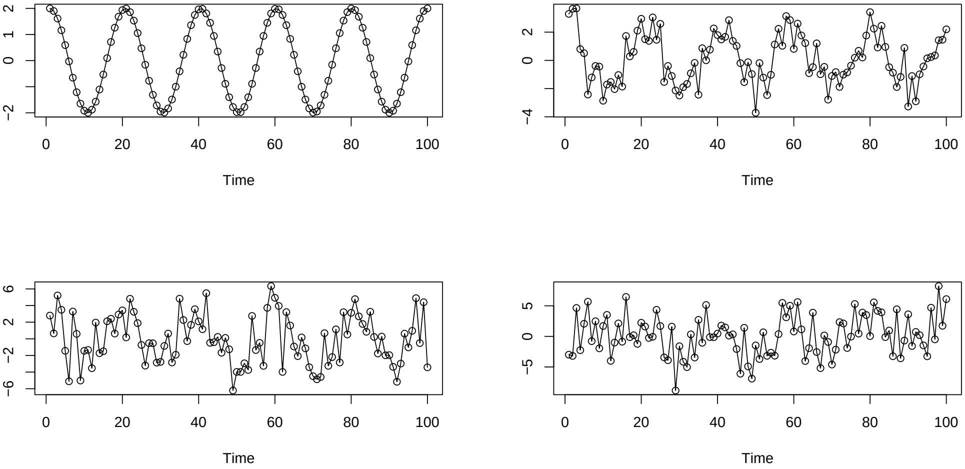
Discussion

- There is no trend in the mean of this random walk process. Recall that $\mu_t = \mathrm{E}(Y_t) = 0$, for all t. However, it would be easy to incorrectly assert that true downward and upward trends are present.

Discussion

- On the other hand, it may be hard to detect trends if the data are very noisy. For example, a noisy realization of a sinusoidal process. It is easy to miss the true cyclical structure from looking at the plot.

Discussion



Deterministic Trend Models

- In this chapter, we consider models of the form

$$Y_t = \mu_t + X_t,$$

  where μ_t is a deterministic function that describes the trend and X_t is random error.

- Note that if, in addition, $\mathrm{E}(X_t) = 0$ for all t, then $\mathrm{E}(Y_t) = \mu_t$ is the mean function for the process $\{Y_t\}$.

- In practice, different deterministic trend functions could be considered.

Estimation of a constant mean

Estimation of a constant mean

- We first consider the most elementary type of trend, namely, a constant trend. Specifically, we consider the model

$$Y_t = \mu + X_t,$$

where μ is constant (free of t) and where $\mathrm{E}(X_t) = 0$. Note that, under this zero mean error assumption, we have

$$\mathrm{E}(Y_t) = \mu.$$

That is, the process $\{Y_t\}$ has an overall population mean function $\mu_t = \mu$, for all t.

Estimation of a constant mean

- The most common estimate of μ is

$$\overline{Y} = \frac{1}{n}\sum_{t=1}^{n} Y_t,$$

  the sample mean.

- It is easy to check that $\overline{Y}$ is an unbiased estimator of μ.

Estimation of a constant mean

- RESULT: If $\{Y_t\}$ is a stationary process with autocorrelation function ρ_k, then
$$\operatorname{Var}(\overline{Y}) = \frac{\lambda_0}{n}\left[1 + 2\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\rho_k\right],$$
where $\operatorname{Var}(Y_t) = \lambda_0$.

- RECALL: If $\{Y_t\}$ is an iid process, that is, $Y_1, Y_2, \ldots, Y_n$ is an iid (random) sample, then
$$\operatorname{Var}_{iid}(\overline{Y}) = \frac{\lambda_0}{n}.$$

Result

Suppose that $Y_t = \mu + X_t$, where μ is constant, $X_t \sim N(0, \lambda_0)$, and $\{X_t\}$ is a stationary process. Under these assumptions, $Y_t \sim N(\mu, \lambda)$ and $\{Y_t\}$ is stationary. Therefore,

$$\overline{Y} \sim N\left(\mu, \frac{\lambda_0}{n}\left[1 + 2\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\rho_k\right]\right)$$

- If λ_0 and the ρ_k's are known, then a $100(1-\alpha)$ percent confidence interval for μ is

$$\overline{Y} \pm z_{\alpha/2}\sqrt{\frac{\lambda_0}{n}\left[1+2\sum_{k=1}^{n-1}\left(1-\frac{k}{n}\right)\rho_k\right]}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ quantile from the standard normal distribution.

Remark

Of course, in real life, rarely will anyone tell us the values of λ_0 and the ρ_k's. These are model (population) parameters. However, if the sample size n is large and good (large-sample) estimates of these quantities can be calculated, we would expect this interval to be approximately valid when the estimates are substituted in for the true values.

Regression methods

Straight line model

- We now consider the deterministic time trend model

$$\begin{aligned} Y_t &= \mu_t + X_t \\ &= \beta_0 + \beta_1 t + X_t, \end{aligned}$$

  where $\mu_t = \beta_0 + \beta_1 t$ and where $\mathrm{E}(X_t) = 0$.

- We are considering a simple linear regression model for the process $\{Y_t\}$, where time t is the predictor.

- By fitting this model, we mean that we would like to estimate the regression parameters β_0 and β_1 (the intercept and slope, respectively) using the observed data $Y_1, Y_2, \ldots, Y_n$.

- The X_t's are random errors and are not observed.

Least Squares Estimation

- To estimate β_0 and β_1, we will use the method of least squares.
- Specifically, we find the values of β_0 and β_1 that minimize the objective function

$$Q(\beta_0, \beta_1) = \sum_{t=1}^{n} (Y_t - \mu_t)^2 = \sum_{t=1}^{n} (Y_t - \beta_0 - \beta_1 t)^2$$

Least Squares Estimation

- This can be done using the partial derivatives of $Q(\beta_0, \beta_1)$

$$\frac{\partial Q(\beta_0, \beta_1)}{\partial \beta_0} = -2\sum_{t=1}^{n}(Y_t - \beta_0 - \beta_1 t), \quad \frac{\partial Q(\beta_0, \beta_1)}{\partial \beta_1} = -2\sum_{t=1}^{n} t(Y_t - \beta_0 - \beta_1 t)$$

- Setting these derivatives equal to zero and jointly solving for β_0, β_1 we get

$$\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{t}, \text{ and } \quad \hat{\beta}_1 = \frac{\sum_{t=1}^{n}(t - \overline{t})\, Y_t}{\sum_{t=1}^{n}(t - \overline{t})^2}.$$

- These are the least squares estimators of β_0, β_1.

Properties

- Under just the mild assumption of $\mathrm{E}(X_t) = 0$, for all t, the least squares estimators are unbiased. That is,

$$\mathrm{E}(\hat{\beta}_0) = \beta_0, \qquad \text{and} \qquad E(\hat{\beta}_1) = \beta_1.$$

- Under the assumptions that $\mathrm{E}(X_t) = 0$, $\{X_t\}$ independent, and $\mathrm{Var}(X_t) = \lambda_0$ (a constant, free of t), then

$$\mathrm{Var}(\hat{\beta}_0) = \lambda_0 \left[\frac{1}{n} + \frac{\bar{t}^2}{\sum_{t=1}^{n} (t - \bar{t})^2} \right], \quad \text{and} \quad \mathrm{Var}(\hat{\beta}_1) = \frac{\lambda_0}{\sum_{t=1}^{n} (t - \bar{t})^2}.$$

- Note that a zero mean white noise process $\{X_t\}$ satisfies these assumptions.

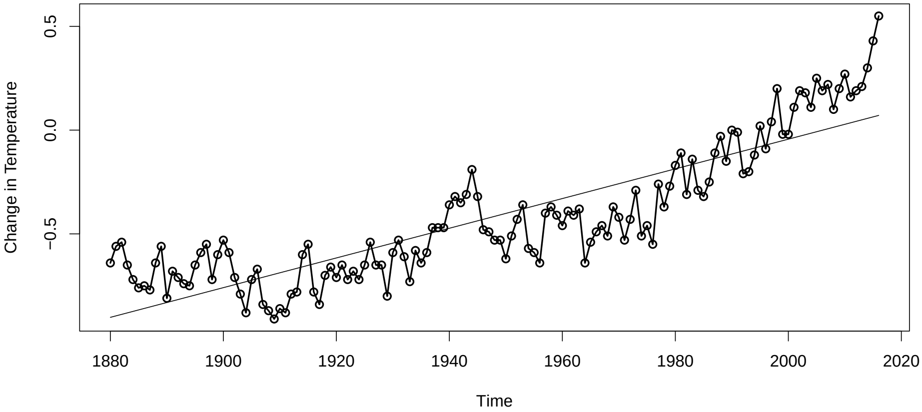
Properties

- In addition to the assumptions $\mathrm{E}(X_t) = 0$, $\{X_t\}$ independent, and $\mathrm{Var}(X_t) = \lambda_0$, if we also assume that the X_t's are normally distributed, then
$$\hat{\beta}_0 \sim N\left(\beta_0, \lambda_0 \left[\frac{1}{n} + \frac{\bar{t}^2}{\sum_{t=1}^{n}(t-\bar{t})^2}\right]\right)$$
and
$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\lambda_0}{\sum_{t=1}^{n}(t-\bar{t})^2}\right)$$

IMPORTANT

- These four assumptions on the errors X_t, that is, zero mean, independence, homoscedasticity, and normality, are the usual assumptions on the errors in a standard regression setting.
- If at least one of these assumptions is violated, then standard errors of the estimators, confidence intervals, t- tests, p-values, etc., will not be meaningful.
- The only instance in which these are exactly true is if $\{X_t\}$ is a zero-mean normal white noise process.

Example Linear Fit



Example Linear Fit: R Output

```
Call:
lm(formula = temps ~ time(temps))

Residuals:
     Min       1Q   Median       3Q      Max 
-0.33885 -0.11682 -0.02592  0.11514  0.47912 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.435e+01  6.860e-01  -20.92   <2e-16 ***
time(temps)  7.154e-03  3.521e-04   20.32   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.163 on 135 degrees of freedom
Multiple R-squared:  0.7536,	Adjusted R-squared:  0.7518 
F-statistic: 412.9 on 1 and 135 DF,  p-value: < 2.2e-16
```

Polynomial regression

- We now consider the deterministic time trend model

$$\begin{aligned} Y_t &= \mu_t + X_t \\ &= \beta_0 + \beta_1 t + \beta_2 t^2 + \ldots + \beta_k t^k + X_t, \end{aligned}$$

where $\mu_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \ldots + \beta_k t^k$ and where $\mathrm{E}(X_t) = 0$. The mean function μ is a polynomial function with degree $k \geq 1$.

Least Square Estimation

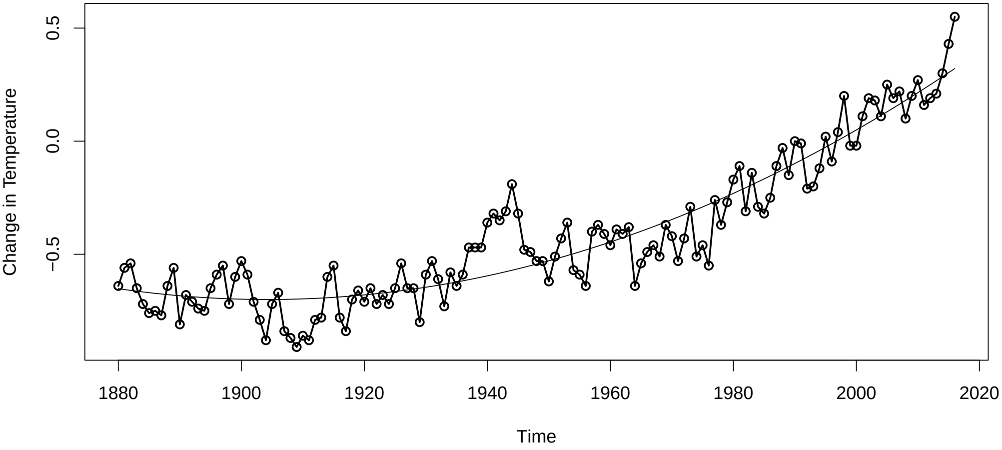
- The least squares estimates of $\beta_0, \ldots, \beta_k$ are obtained in the same way as in the $k = 1$ case; namely, the estimates are obtained by minimizing the objective function

$$Q(\beta_0, \ldots, \beta_k) = \sum_{t=1}^{n} (Y_t - (\beta_0 + \beta_1 t + \beta_2 t^2 + \ldots + \beta_k t^k))^2$$

with respect to $\beta_0, \ldots, \beta_k$

- There are no closed-form expressions for the least squares estimators when $k > 1$. We use numerical methods and computing to fit the model.
- Under the mild assumption that the errors have zero mean; it follows that the least squares estimators are unbiased estimators of their population analogues.
- As in the simple linear regression case $(k = 1)$, additional assumptions on the errors are needed to derive the sampling distribution of the least squares estimators, namely, independence, constant variance, and normality.
- Regression output (e.g., in R, etc.) is correct only under these additional assumptions.

Example Quadratic Fit



Example Quadratic Fit: R Output

```
Call:
lm(formula = temps ~ time + time2)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27095 -0.08627  0.00622  0.07469  0.38076 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.957e+02  2.682e+01   11.03   <2e-16 ***
time        -3.113e-01  2.754e-02  -11.30   <2e-16 ***
time2        8.173e-05  7.069e-06   11.56   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1157 on 134 degrees of freedom
Multiple R-squared:  0.8767,	Adjusted R-squared:  0.8748 
F-statistic: 476.2 on 2 and 134 DF,  p-value: < 2.2e-16
```

Seasonal means model

Consider the deterministic trend model where the mean function

$$\mu_t = \begin{cases} \beta_1, & t = 1, 13, 25, \ldots \\ \beta_2, & t = 2, 14, 26, \ldots \\ \vdots, & \vdots \\ \beta_{12}, & t = 12, 24, 36, \ldots \end{cases}$$

The regression parameters $\beta_1, \beta_2, \ldots, \beta_{12}$ are fixed constants. This is called a seasonal means model.

Seasonal means model

- This model does not take the shape of the seasonal trend into account; instead, it merely says that observations 12 months apart have the same mean, and this mean does not change through time.

- Other seasonal means models with a different number of parameters could be specified. For instance, for quarterly data, we could use a mean function with 4 regression parameters $\beta_1, \beta_2, \beta_3, \beta_4$

Fitting the model

We can still use least squares to fit the seasonal means model. The least squares estimates of the regression parameters are simple to compute, but difficult to write mathematically. In general,

$$\hat{\beta}_i = \frac{1}{n_i} \sum_{t \in \mathcal{A}_i} Y_t$$

where the set $\mathcal{A}_i = \{t : t = i + 12j, j = 0, 1, 2, \ldots, \}$, for $i = 1, 2, ..., 12$, where n_i is the number of observations in month i.

Cosine trend model

Consider the deterministic time trend model

$$
\begin{aligned}
Y_t &= \mu_t + X_t \\
&= \beta \cos(2\pi f t + \phi) + X_t
\end{aligned}
$$

The trigonometric mean function μ_t consists of different parts:

- β is the amplitude. The function μ_t oscillates between $-\beta$ and β.
- f is the frequency. $1/f$ is the period (the time it takes to complete one full cycle of the function). For monthly data, the period is 12 months; i.e., the frequency is $f = 1/12$.
- ϕ controls the phase shift. This represents a horizontal shift in the mean function.

Model Fitting

- Fitting this model is difficult unless we transform the mean function into a simpler expression. We can rewrite Y_t as

$$Y_t = \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + X_t,$$

  where $\beta_1 = \beta \cos(\phi)$ and $\beta_2 = -\beta \sin(\phi)$.

- Thus, Y_t is a linear function of β_1 and β_2, where $\cos(2\pi f t)$ and $\sin(2\pi f t)$ play the roles of predictor variables.

Remarks

- The seasonal means and cosine trend models are competing models; that is, both models are useful for seasonal data.
- The cosine trend model is more parsimonious; i.e., it is a simpler model because there are 3 regression parameters to estimate. On the other hand, the (monthly) seasonal means model has 12 parameters that need to be estimated!
- The seasonal means model does not require a specific shape for the periodicity.

Interpreting Regression Output

Coefficient of Determination

- If $\mathrm{Var}(X_t) = \gamma_0$ is constant, an estimate of γ_0 is given by

$$S^2 = \frac{1}{n-p} \sum_{t=1}^{n} (Y_t - \hat{\mu}_t)^2,$$

  where $\hat{\mu}_t$ is the least squares estimate of μ_t and p is the number of regression parameters in μ_t.

- The term $n - p$ is called the error degrees of freedom.

Coefficient of Determination

- For any data set Y_t we can write algebraically

$$\underbrace{\sum_{t=1}^{n}(Y_t - \overline{Y})^2}_{\text{SST}} = \underbrace{\sum_{t=1}^{n}(\hat{Y}_t - \overline{Y})^2}_{\text{SSR}} + \underbrace{\sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2}_{\text{SSE}}.$$

- These quantities are called sums of squares and form the basis for the analysis of variance (ANOVA).

- Since $SST = SSR + SSE$, it follows that the proportion of the total variation in the data explained by the deterministic model is

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST},$$

  the coefficient of determination.

- The larger R^2 is, the better the deterministic part of the model explains the variability in the data.

Remarks

- R^2 is computed under the assumption that the deterministic trend model is correct and assesses how much of the variation in the data may be attributed to that relationship rather than just to inherent variation.
- If R^2 is small, it may be that there is a lot of random inherent variation in the data, so that, although the deterministic trend model is reasonable, it can only explain so much of the observed overall variation.
- Alternatively, R^2 may be close to 1, but a particular model may not be the best model. In fact, R^2 could be very high, but not relevant because a better model may exist.

Residual Analysis (Model Diagnostics)

Residuals

- The fitted model is $\hat{Y}_t = \hat{\mu}_t$ and the residual process is
$$\hat{X}_t = Y_t - \hat{Y}_t.$$
- The residuals from the model fit are important. In essence, they serve as proxies (predictions) for the true errors X_t, which are not observed.
- The residuals can help us learn about the validity of the assumptions made in our model.

Standardized Residuals

- If the model above is fit using least squares, then the sum of the residuals is equal equal zero. Thus, the residuals have mean zero and the standardized residuals, defined by

$$\hat{X}_t^* = \frac{\hat{X}_t}{S}.$$

Assessing normality

- We examine the (standardized) residuals and look for evidence of normality.
- We can use histograms and normal probability plots (also known as quantile-quantile, or qq plots) to do this.

Assessing normality: Shapiro-Wilk Test

- Histograms and qq plots provide only visual evidence of normality.
- The Shapiro-Wilk test is a formal hypothesis test that can be used to test

  $$H_0 : \text{the (standardized) residuals are normally distributed}$$

  versus

  $$H_1 : \text{the (standardized) residuals are not normally distributed.}$$

- The test is carried out by calculating a statistic W approximately equal to the sample correlation between the ordered (standardized) residuals and the normal scores.

Assessing Independence

- Plotting the residuals versus time can provide visual insight on whether or not the (standardized) residuals exhibit independence (although it is often easier to detect gross violations of independence).
- Residuals that hang together are not what we would expect to see from a sequence of independent random variables.
- Similarly, residuals that oscillate back and forth too notably also do not resemble this sequence.

Assessing Independence: Runs Test

- A runs test is a nonparametric test which calculates the number of runs in the (standardized) residuals. The formal test is

$$H_0 : \text{the (standardized) residuals are independent}$$

versus

$$H_1 : \text{the (standardized) residuals are not independent.}$$

Assessing Independence: Runs Test

In particular, the test examines the (standardized) residuals in sequence to look for patterns that would give evidence against independence. Runs above or below 0 (the approximate median of the residuals) are counted.

- A small number of runs would indicate that neighboring values are positively dependent and tend to hang together over time.
- Too many runs would indicate that the data oscillate back and forth across their median. This suggests that neighboring residuals are negatively dependent.
- Therefore, either too few or too many runs lead us to reject independence.

Sample Autocorrelation Function

- For a set of time series data $Y_1, Y_2, \cdots, Y_n$, we define the sample autocorrelation function, at lag k, by

$$r_k = \frac{\sum_{t=k+1}^{n}(Y_t - \overline{Y})(Y_{t-k} - \overline{Y})}{\sum_{t=1}^{n}(Y_t - \overline{Y})^2}$$

  where $\overline{Y}$ is the sample mean of $Y_1, Y_2, \cdots, Y_n$.

- The sample version r_k is a point estimate of the true (population) autocorrelation ρ_k.

Sample Autocorrelation Function

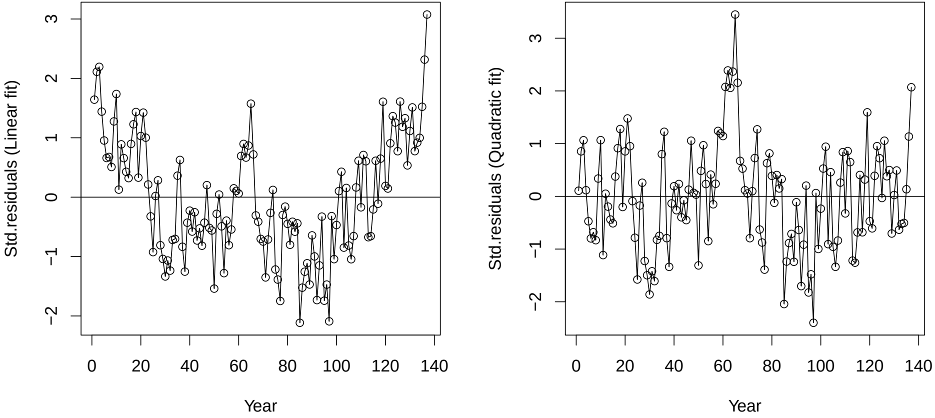
- If the standardized residual process $\hat{X}_t^*$ is white noise, then for n large

$$r_k \sim \mathcal{N}\left(0, \frac{1}{n}\right).$$

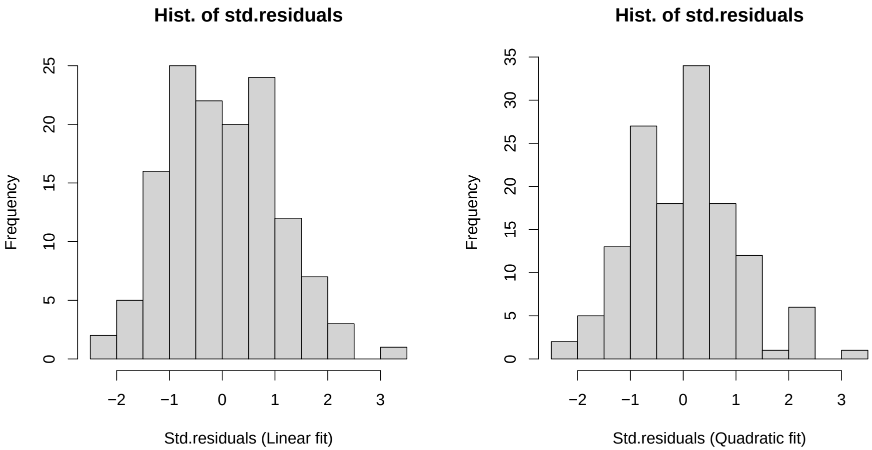
- Thus, we construct a confidence interval to test the hypothesis that the sample autecorrelations are null for all k. This interval is

$$\pm 2/\sqrt{n}$$

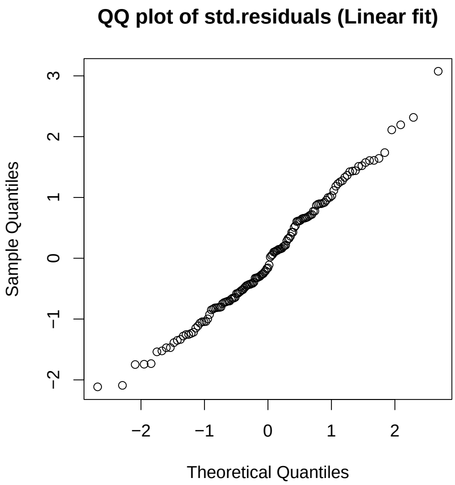
Residual Analysis of Linear and Quadratic fit for Global Temperature

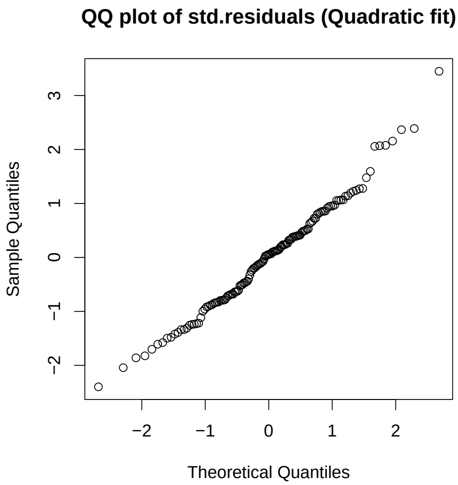


Residual Analysis of Linear and Quadratic fit for Global Temperature

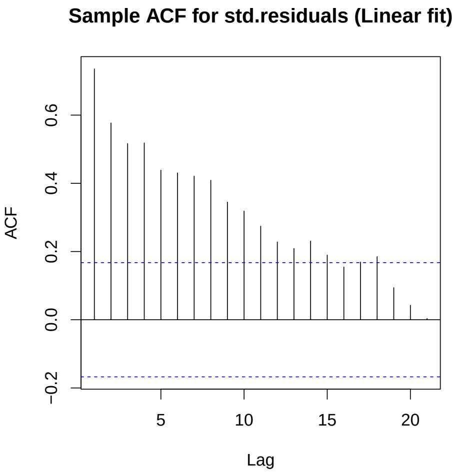


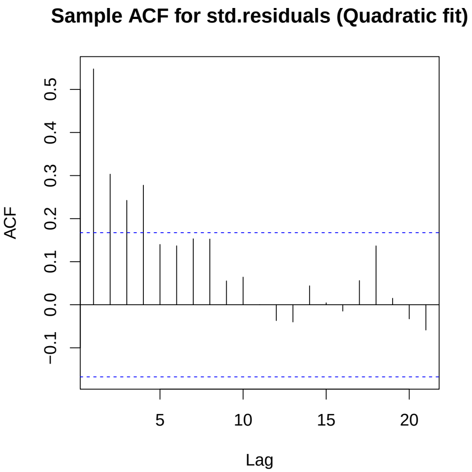
Residual Analysis of Linear and Quadratic fit for Global Temperature





Residual Analysis of Linear and Quadratic fit for Global Temperature





Residual Analysis of Linear and Quadratic fit for Global Temperature

```
> shapiro.test(rstudent(Linear_fit))

	Shapiro-Wilk normality test

data:  rstudent(Linear_fit)
W = 0.98604, p-value = 0.1791

> runs(rstudent(Linear_fit))
$pvalue
[1] 4.06e-15

$observed.runs
[1] 25

$expected.runs
[1] 69.46715

$n1
[1] 70

$n2
[1] 67

$k
[1] 0
```

```
> shapiro.test(rstudent(Quadratic_fit))

	Shapiro-Wilk normality test

data:  rstudent(Quadratic_fit)
W = 0.98787, p-value = 0.2733

> runs(rstudent(Quadratic_fit))
$pvalue
[1] 0.0172

$observed.runs
[1] 55

$expected.runs
[1] 69.32117

$n1
[1] 65

$n2
[1] 72

$k
[1] 0
```